

Research Highlights

- **Highlight Title 1**
One-line description of this research highlight.
[\[website\]](#) [\[code\]](#)
- **Highlight Title 2**
One-line description of this research highlight.
[\[website\]](#) [\[code\]](#)
- **Highlight Title 3**
One-line description of this research highlight.
[\[website\]](#) [\[code\]](#)

Education

- **The University of Hong Kong** Hong Kong SAR
Ph.D. candidate at IDS advised by Prof. Yi Ma Sep 2024 - Now
- **ShanghaiTech University** Shanghai, China
B.Eng in Computer Science and Technology (Honor) Sep 2020 - Jun 2024
- **University of California, Berkeley** Berkeley, CA, US
Visiting Undergraduate, EECS & BAIR Aug 2022 - May 2023

Publications

(* means equal contribution)

- *SFT Memorizes, RL Generalizes: A Comprehensive Study of Foundation Model Post-training*
Tianzhe Chu*, Yuexiang Zhai*, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, Yi Ma
ICML 2025 [\[paper\]](#) [\[website\]](#) [\[code\]](#)
- *Pointer-CAD: Unifying B-Rep and Command Sequences via Pointer-based Edges & Faces Selection*
Dacheng Qi, Chenyu Wang, Jingwei Xu, **Tianzhe Chu**, Zibo Zhao, Wen Liu, Wenrui Ding, Yi Ma, Shenghua Gao
CVPR 2026 [\[paper\]](#)
- *Seeing from Another Perspective: Evaluating Multi-View Understanding in MLLMs*
Chun-Hsiao Yeh*, Chenyu Wang*, Shengbang Tong, Ta-Ying Cheng, Ruoyu Wang, **Tianzhe Chu**, Yuexiang Zhai, Yubei Chen, Shenghua Gao, Yi Ma
AAAI 2026 [\[paper\]](#) [\[website\]](#) [\[code\]](#)
- *White-Box Transformers via Sparse Rate Reduction: Compression Is All There Is?*
Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Hao Bai, Yuexiang Zhai, Benjamin D. Haeffele, Yi Ma
JMLR [\[paper\]](#)
- *Image Clustering via the Principle of Rate Reduction in the Age of Pretrained Models*
Tianzhe Chu*, Shengbang Tong*, Tianjiao Ding*, Xili Dai, Benjamin D. Haeffele, René Vidal, Yi Ma
ICLR 2024 [\[paper\]](#) [\[code\]](#)
- *Emergence of Segmentation with Minimalistic White-Box Transformers*
Yaodong Yu*, **Tianzhe Chu***, Shengbang Tong, Ziyang Wu, Druv Pai, Sam Buchanan, Yi Ma
CPAL 2024 (Oral), **NeurIPS 2023 XAI Workshop (Oral)** [\[paper\]](#) [\[website\]](#) [\[code\]](#)
- *White-Box Transformers via Sparse Rate Reduction*
Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Benjamin D. Haeffele, Yi Ma
NeurIPS 2023 [\[paper\]](#) [\[website\]](#) [\[code\]](#)
- *GenDexHand: Generative Simulation for Dexterous Hands*
Feng Chen*, Zhuxiu Xu*, **Tianzhe Chu**, Xunzhe Zhou, Li Sun, Zewen Wu, Shenghua Gao, Zhongyu Li, Yanchao Yang, Yi Ma
Preprint [\[paper\]](#)

- [GAUSS Eval: Human-LLM Judge Consistency Analysis](#)
Tianzhe Chu*, Jiaxin Zhang*, Zhenyu Liao, Qiuyu Ren, Tahsin Saffat, Zitong Yang, Yi Ma, Yue Zhang*
Blog Post [\[website\]](#)

Open Source

- **Deep Representation Learning Book** – Core Contributor [\[code\]](#)
Open-source textbook on the principles of deep representation learning.
- **CRATE** – Core Contributor [\[code\]](#)
White-box transformer architecture derived from sparse rate reduction.

Selected Awards

- **Outstanding Graduate** Shanghai, China
Jul, 2024
Affiliation: ShanghaiTech University
- **Merit Student** Shanghai, China
Dec 2023
Affiliation: ShanghaiTech University

Invited Talks

- Localize the Credits of Reasoning in Roaring 2025: LLM Reasoning Paper Potpourri @ ALICE, Aug. 2025
- SFT Memorizes, RL Generalizes @ BAAI, Jun. 2025
- SFT Memorizes, RL Generalizes @ Cathy Wu's lab, MIT, Apr. 2025
- White-Box Transformers @ SVIP Lab, ShanghaiTech, Sep. 2023
- Emergence of Segmentation in White-Box Transformers @ CPAL 2024, HKU, Jan. 2024

Service

- **Head Teaching Assistant**
HKU DATA 8014
- **Reviewer**
ICLR 2025/2026, ICML 2025, NeurIPS 2025, CVPR 2026